

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

II Semester of M. Sc. (AOC) Examination May 2018

OC 729 Fundamentals of Physical Chemistry-II

Date: 04/05/2018 Day: Friday Time: 01.30 pm To 02.00 pm Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I Choose the correct answer for the following questions.

[20]

1. It is only the absorbed light radiations that are effective in producing a chemical reaction." This is the statement of
(a) Lambert law (b) Lambert-Beer law (c) Grothus-Draper law (d) Stark-Einstein law ☐
2. With rise in temperature, the surface tension of a liquid is _____
(a) increases (b) decreases (c) remains the same (d) none of the above ☐
3. A species which can both absorb and transfer radiant energy for activation of the reactant molecule is called
(a) radioactive substance (b) an ionizer (c) a photochemical substance (d) a photosensitizer ☐
4. For a orthorhombic crystal system, which is incorrect?
(a) $a \neq b \neq c$ (b) $\alpha = \beta = \gamma = 90^\circ$ (c) $a = b \neq c$ (d) none of these ☐
5. The formula used for the determination of surface tension by capillary rise method is
(a) $2\gamma = hrdg$ (b) $2\gamma = hr2dg$ (c) $2\gamma = \pi r \cos \theta$ (d) $2\gamma = \pi hr2dg$ ☐
6. An intensive property does not depend upon
(a) nature of the substance (b) quantity of matter (c) external temperature (d) atmospheric pressure ☐

7. If there are 4 atoms in unit cell in a cubic system, it is an example of
 (a) simple cubic unit cell (b) body centred cubic unit cell ☐
 (c) face centred cubic unit cell (d) none of these
8. The dipole moment of a polar substance is given by the formula, where q is charge at one end and r is the distance between the opposite charges
 (a) $\mu = q + r$ (b) $\mu = q - r$ (c) $\mu = q \times r$ (d) $\mu = q \div r$ ☐
9. In Bragg's equation $n \lambda = 2 d \sin \theta$, ' n ' represents
 (a) the number of moles (b) the principal quantum number ☐
 (c) the Avogadro's number (d) the order of reflection
10. The emission of light as a result of chemical action is called _____
 (a) phosphorescence (b) fluorescence (c) chemiluminescence (d) none of these ☐
11. The units erg, joule and calorie are interconvertible, which of the following is incorrect?
 (a) $107 \text{ ergs} = 1 \text{ Joule}$ (b) $4.184 \text{ J} = 1 \text{ cal.}$ (c) $1 \text{ Joule} = 0.2390 \text{ cal.}$ (d) $1 \text{ erg} = 4.184 \text{ cal.}$ ☐
12. The SI unit of coefficient of viscosity is _____
 (a) $\text{kg m}^2 \text{ sec}$ (b) kg m sec^{-1} (c) $\text{kg m}^{-1} \text{ sec}$ (d) $\text{kg m}^{-1} \text{ sec}^{-1}$ ☐
13. Which is the correct unit for entropy?
 (a) kJ mole (b) $\text{JK}^{-1} \text{ mole}$ (c) $\text{JK}^{-1} \text{ mole}^{-1}$ (d) kJ mole ☐
14. The example of elastomer is _____
 (a) PVC (b) PS (c) silicone rubber (d) cotton ☐
15. The number of molecules reacted or formed per photon of light absorbed is called
 (a) yield of the reaction (b) quantum efficiency (c) quantum yield (d) quantum productivity ☐
16. If intrinsic viscosity is $200 \text{ cm}^3/\text{g}$ and reduced viscosity is 0.5 then concentration is _____
 (a) 0.002 g/ml (b) 0.0025 g/ml (c) 0.005 g/ml (d) 0.5 ☐
17. Viscosity of a liquid is a measure of
 (a) repulsive forces between the liquid molecules (b) frictional resistance ☐
 (c) intermolecular forces between the molecules (d) none of the above
18. Potassium crystallizes in a bcc structure. The co-ordination number of potassium in potassium metal is
 (a) 2 (b) 4 (c) 6 (d) 8 ☐
19. In Syndiotactic Polymer the arrangement of groups in chain are
 (a) Same side (b) Random Position (c) Alternate side (d) cross linked ☐
20. The reactions which are caused by heat and in absence of light are called
 (a) photochemical reactions (b) catalytic reactions ☐
 (c) exothermic reactions (d) thermal or dark reactions

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

II Semester of M. Sc. (AOC) Examination May 2018

OC 729 Fundamentals of Physical Chemistry-II

Date: 04/05/2017 Day: Friday Time: 02.00 pm To 04.30 pm Maximum Marks: 50

Instructions to the candidates:

1. **Section-I** and **Section-II** must be attempted in **Two answer sheet**.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of **non-programmable calculator** is **allowed**.
4. Show necessary calculation.

SECTION-I [20]

Q-II

- [A] Derive the equation for Lewis Randall Rule. [05]

OR

- [A] Derive the equation $\frac{f}{p} = \frac{P}{P_{id}}$ and For a gas at 0°C $PV = RT (1 - 9.93 \times 10^{-4} P + 2.19 \times 10^{-6} P^2)$ [05]

calculate the fugacity at 10 atm. pressure

- [B] Calculate the fugacity of 1 mole of gas X at 25°C and 200 atm. The integral value [05]

$\int_0^P (V - \frac{RT}{P}) dp$ is - 20.02 at 200 atm. Assuming that free energy of gas X at 25°C and 1 atm is - 12 Kcal/mole. Calculate the free energy of X at 25°C and 200 atm.

Q-III

- [A] Derive the equation for catalyzed and non-catalyzed poly condensation. [06]

- [B] Discuss the lowering of vapour pressure method for determination mol. wt. of polymer. [04]

OR

- [B] List out the different method for determination of mol. wt. of polymer based on colligative [04]
properties and discuss the end group analysis method.

SECTION-II [30]

Q-IV

- [A] Write the differences between fluorescence and phosphorescence. [05]

- [B] What is photosensitization and quenching? Discuss its mechanism with suitable example. [05]

OR

- [B] Derive the equation for Lambert and beer's law. Explain quantum yield and derive its unit. [05]

Q-V

- [A] What is viscosity? Discuss the method for determination of viscosity and explain effect of [04]
temperature on viscosity.

OR

[A] Discuss the capillary rise method for determination of surface tension and also discuss the effect of temperature on surface tension. [04]

[B] Differentiate between homo diatomic molecule and hetero diatomic molecule. [03]

[C] What is dipole moment? Derive unit of dipole moment. [03]

Q-VI

[A] For LiBH_4 crystallize in orthorhombic system. The unit cell dimensions are $a = 6.81 \text{ \AA}$, $b = 4.4 \text{ \AA}$ and $c = 7.2 \text{ \AA}$. If mol. wt. is 21.7 g/mole and density is 0.67 g/ml . Find out no. of atom per unit cell and identify lattice type. [04]

[B] Obtain the atom per unit cell, coordination number and packing fraction for SC, BCC and FCC. [06]

OR

[B] What is miller indices? Discuss its physical significance. Draw the crystal structure for (100), (010) and (001). [06]

Useful Physical Parameters:

Planck's constant, $h = 6.626068 \times 10^{-34} \text{ J.s}$

Universal gas constant, $R = 8.31 \text{ J/mol. K}$

Avogadro number, $N_A = 6.023 \times 10^{23}$

Boltzmann Constant, $k = 1.38 \times 10^{-23} \text{ J/K}$

Speed of light in vacuum, $c = 3.0 \times 10^8 \text{ m/s}$